# Yusuf Miraç Uyar

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## Education

**Technical University of Munich** — Munich, Germany
*M.Sc. in Informatics* — *Oct. 2023 – Present*
**Current grade: 1.8** *(German grading scale; 1.0 = best)*

**Bilkent University** — Ankara, Turkey
*B.S. in Computer Engineering, Full Merit Scholarship* — *Sep. 2018 – 2023*
**GPA: 3.64 / 4.00**

## Research Projects

**Multi-Agent Reinforcement Learning for Flying Ad-Hoc Networks** — Aug. 2026 – Present
*Master's Thesis, Fraunhofer IKS* — *Prof. Martin Bichler*

- Developed a **permutation-equivariant transformer policy** with sequential partner selection for decentralized optical aircraft networks in a use case introduced by a major aerospace company.
- Designed difference rewards for multi-agent credit assignment and compared **MAPPO/IPPO**, negotiation protocols, and synthetic versus real-flight training data.
- Built centralized optimization benchmarks using graph reachability and mixed-integer linear programming to quantify connectivity limits; evaluated on a full day of European flight data.
- On the **24-hour evaluation day**, the best learned policy achieved **90% of the nighttime connectivity upper bound**, compared with **73%** for an industry-method baseline, with comparable link stability.
- Currently investigating how aircraft density affects achievable connectivity and the traffic density required to maintain reliable paths to ground stations.

**Digital Platform Detector** — Winter 2025/26
*Interdisciplinary Project, TUM* — *Dr. David Angenendt*

- Fine-tuned **Mistral-7B with LoRA** on ORBIS/Crunchbase descriptions; achieved **0.82 F1** on 116 validation cases using literature-derived and expert-labelled training data.
- Constructed a **735-company dataset** by matching literature-derived platform companies to ORBIS, enriching descriptions with Crunchbase, and incorporating expert-labelled boundary cases.
- Refined classification instructions and negative examples, reducing the false-positive rate from **34.1% to 11.4%** and increasing precision from **74.1% to 91.4%** on the same **116 validation cases**.

**One Out Training for Physics Emulators Using Continuous Convolutions** — Summer 2025
*Guided Research, TUM* — *Prof. Nils Thuerey*

- Evaluated deep learning for particle-based physical simulation across **three fluid datasets**, comparing continuous-convolution kernel bases, message-passing depth, and accuracy–compute trade-offs.
- Adapted LagrangeBench TGV-2D trajectories to the SFBC pipeline, computing missing SPH density fields, particle areas, and support radii while preserving periodic boundaries.
- Developed a **hybrid MSE–correlation loss** to improve the visual fidelity of predicted density fields. In a LagrangeBench TGV-2D comparison, reduced density-prediction MSE by **94.4%** and increased spatial correlation from **0.955 to 0.999**, without extra inference cost.

**Benchmarking Provably Safe Reinforcement Learning** — Winter 2024/25
*Master's Practicum, TUM* — *Prof. Matthias Althoff*

- Implemented **zonotope-based action-set representations**; investigated how geometry and dimensionality affect **PPO** with continuous action masking and action replacement in MuJoCo.
- Compared L1- and L2-based action constraints for continuous robotic control, examining how their geometry affects the admissible actions and the computational cost of safe exploration.
- Worked with **quadrotor dynamics and state-dependent safety constraints** through the team's 2D and 3D simulation benchmarks, examining how action representations affect safe reinforcement learning.
- Analysed how increasing action-space dimensionality restricts feasible control inputs and raises computational costs, motivating more scalable L2-based zonotope representations.

## Experience

**Fraunhofer IKS** Garching, Germany
*Master's Thesis Research* *Aug. 2026 – Present*
- Exploring classical reinforcement learning for flying ad hoc networks within a project comparing classical and quantum-based reinforcement learning approaches.

**Olympos** Istanbul, Turkey
*Data Scientist – Part-Time* *June 2024 – Jan. 2026*
- Built **LLM**-based models to analyze companies and predict VC firms' selection behavior with **73%** accuracy.
- Built early-stage **Y Combinator** detection models by combining **LLMs** with LinkedIn profile analysis.
- Enhanced a salary prediction pipeline by improving **data integration** across Glassdoor, Levels.fyi, and Crunchbase through a **Python**-based seniority inference model and **multithreaded processing**.

*Data Scientist – Part-Time* *June 2022 – May 2023*
- Developed a tokenization method in **Python** and **Pandas** that reduced Google Places API usage by **32%**.
- Identified and corrected **3%** of database record errors related to bias and network issues using **PostgreSQL**.
- Developed and deployed company quantification analyses using LinkedIn profile data.

**Bilkent University** Ankara, Turkey
*Undergraduate Research Assistant* *July 2021 – Mar. 2023*
- Researched StyleGAN architectures for kinship face generation in **PyTorch**, supervised by Dr. Hamdi Dibeklioğlu.

**SBI Bilişim A.S.** Ankara, Turkey
*Software Engineering Intern* *June 2020 – September 2020*
- Implemented functionalities to visualize trained convolutional neural networks using **Keras** and **TensorFlow**.
- Used those methods to combine trained networks that improved object detection by **8%** in the final product.

## Bachelor Projects

**Tutorium** Fall 2022
*TypeScript, React, HTML, CSS, SemanticUI*
- Developed a platform that matched students with tutors who excelled in their subjects.
- Built the UI and backend integration using **TypeScript**, **React**, **CSS**, and **SemanticUI**.
- Information Page

**BilHub** Fall 2021
*JavaScript, React, HTML, CSS, SemanticUI*
- Developed a group project management application to organize teams, manage tasks, and review artifacts.
- Designed and developed the application's UI using **CSS** and **SemanticUI**.
- Developed application functionality and database connection using **JavaScript** and **React**.
- GitHub Page

## Skills

**Programming**: Python, C/C++, SQL
**ML**: PyTorch, MARLlib, Gymnasium, Tianshou, scikit-learn
**Tools**: Git, PostgreSQL
**Languages**: English (Proficient), Turkish (Native), German (Beginner)

## Academic Achievements

**Turkish National Olympiad of Informatics** – Bronze Medal 2017